CPTG: Compact-Quiescent Redshift Stress Test of NMBS-C7447 at $z = 1.80$

Carter L Glass Jr

Abstract

NMBS-C7447 is a compact massive quiescent galaxy observed at redshift $z = 1.80$. The published X-Shooter measurement gives an integrated stellar velocity dispersion of $294 \pm 51\,\mathrm{km\,s^{-1}}$, an effective radius of $1.64 \pm 0.15\,\mathrm{kpc}$, a dynamical mass of $(1.7 \pm 0.5) \times 10^{11} M_{\odot}$, and a stellar mass of order $1.5 \times 10^{11} M_{\odot}$. This article uses the object as a compact-quiescent CPTG redshift stress test. The effective-radius support scale is computed from the observed stellar dispersion and radius, then divided by the expansion factor at the observed epoch. The midpoint calculation gives a local support scale of $1.7080 \times 10^{-9}\,\mathrm{m\,s^{-2}}$, which normalizes to $6.1958 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ under the Planck-consistent expansion normalization. The result supports the CPTG time-scaled compact-quiescent acceleration branch in the limited effective-radius sense. It is not a full radial dynamical validation, because the available data do not provide a resolved stellar kinematic field.

1 Motivation

Compact quiescent galaxies at high redshift provide a direct way to test whether a local stellar support scale changes coherently with cosmic time. These systems are useful because their stellar bodies are small, massive, and dynamically hot. Their observed velocity dispersions and effective radii therefore define a compact acceleration scale without requiring a resolved rotation curve.

NMBS-C7447 is a particularly useful target to test Curvature Polarization Transport Gravity (CPTG) [3, 4]. The source paper reports a high-quality VLT X-Shooter spectrum for NMBS-C7447 and gives the compact-galaxy measurements used below, including the stellar velocity dispersion, effective radius, dynamical mass, and stellar mass scale [1]. The measured redshift is

$$z = 1.80. \tag{1}$$

The object was used in the original study to confirm the evolution of the mass-size and mass-dispersion relations for quiescent galaxies. The same published measurements allow a narrow CPTG stress test: compute the effective-radius support scale, normalize it by the expansion factor at the observed epoch, and determine whether the result lies on a plausible present-equivalent compact-galaxy branch.

This paper is intentionally limited. It does not attempt a full radial mode reconstruction, a full Jeans solution, or a lensing calculation. It asks a single redshift-stress question: whether the high local acceleration scale of the compact galaxy at the observed epoch reduces to a stable present-equivalent scale after expansion normalization.

2 Observed Inputs

The calculation uses only the scalar quantities required for a compact effective-radius redshift stress test. The observed inputs are summarized in Table 1 and are taken from the published NMBS-C7447 X-Shooter analysis [1].

Observed quantity	Adopted value	Role in stress test
Galaxy	NMBS-C7447	Target identification
Redshift	$z = 1.80$	Observed epoch and expansion normalization
Stellar velocity dispersion	$294 \pm 51\,\mathrm{km\,s^{-1}}$	Primary pressure-supported anchor
Effective radius	$1.64 \pm 0.15\,\mathrm{kpc}$	Compact effective-radius scale
Dynamical mass	$(1.7 \pm 0.5) \times 10^{11} M_{\odot}$	Independent compact-mass comparator
Stellar mass	$\sim 1.5 \times 10^{11} M_{\odot}$	Photometric mass consistency check

Table 1: Observed inputs used in the compact-quiescent CPTG redshift stress test of NMBS-C7447.

The source paper reports that the dynamical mass and photometric stellar mass are in good agreement [1]. This is useful for the present calculation because it reduces the likelihood that the compact effective-radius support scale is being anchored to an obviously inconsistent stellar mass.

3 Compact Effective-Radius Support Scale

The local support scale is computed from the measured stellar velocity dispersion and effective radius:

$$a_{\sigma} = \frac{\sigma_{\star}^{2}}{R_e}. \tag{2}$$

Using

$$\sigma_{\star} = 294\,\mathrm{km\,s^{-1}}, \tag{3}$$

and

$$R_e = 1.64\,\mathrm{kpc}, \tag{4}$$

gives

$$R_e = 5.061 \times 10^{19}\,\mathrm{m}, \tag{5}$$

and therefore

$$a_\sigma = 1.7080 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{6}$$

This is the observed compact effective-radius support scale at the emission epoch. It is not yet the present-equivalent CPTG stress-test scale because the object is observed at high redshift.

4 Expansion Factor and Cosmic Time

The comparison clock is normalized with the Planck 2018 base-ΛCDM Hubble constant [2]. Thus H_0 is treated explicitly as the Planck-measured Hubble constant used for the dimensional time conversion:

$$H_0^{\mathrm{Planck}} = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{7}$$

The dimensionless expansion-shape parameters are also taken from the same Planck base-ΛCDM reference layer:

$$\Omega_m^{\mathrm{Planck}} = 0.315 \pm 0.007, \tag{8}$$

and

$$\Omega_\Lambda^{\mathrm{Planck}} \simeq 0.685. \tag{9}$$

The expansion factor is

$$E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}. \tag{10}$$

In this expression, Ω_m and Ω_Λ denote the Planck reference values listed above.

The corresponding dimensional Hubble rate is

$$H(z) = H_0^{\mathrm{Planck}} E(z). \tag{11}$$

At the observed redshift,

$$E(1.80) = 2.7568, \tag{12}$$

so that

$$H(1.80) = 185.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{13}$$

The corresponding cosmic age is

$$t(z = 1.80) = 3.62\,\mathrm{Gyr}, \tag{14}$$

with a lookback time of

$$t_{\mathrm{lookback}} = 10.18\,\mathrm{Gyr}. \tag{15}$$

Thus the galaxy is observed when the Universe was approximately 26% of its present age in this reference cosmology.

5 Redshift-Normalized CPTG Stress Scale

The CPTG redshift stress scale is defined by normalizing the local compact support scale by the expansion factor at the observed epoch, following the CPTG compact effective-radius stress-test convention [3, 4]:

$$a_{\sigma,z} = \frac{a_\sigma}{E(z)}. \tag{16}$$

Substituting the midpoint values gives

$$a_{\sigma,z} = \frac{1.7080 \times 10^{-9}}{2.7568} = 6.1958 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{17}$$

In the stress-test interpretation, the local emission-time branch is

$$a_\star(t_{\mathrm{em}}) = a_\sigma, \tag{18}$$

while the present-equivalent branch is

$$a_{\star,0} = a_{\sigma,z}. \tag{19}$$

Therefore

$$a_\star(t_{\rm em}) = 1.7080 \times 10^{-9}\,\mathrm{m\,s^{-2}}, \tag{20}$$

and

$$a_{\star,0} = 6.1958 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{21}$$

The corresponding CPTG curvature radius is tied directly to the same acceleration scale in the CPTG curvature-transport normalization [3]:

$$R_C(t) = \frac{48\pi c^2}{a_\star(t)}. \tag{22}$$

The present-equivalent curvature radius is therefore

$$R_{C,0} = 2.1874 \times 10^{28}\,\mathrm{m}, \tag{23}$$

while the emission-time curvature radius is

$$R_C(t_{\rm em}) = 7.9347 \times 10^{27}\,\mathrm{m}. \tag{24}$$

The higher early-time acceleration branch therefore corresponds to a smaller curvature radius, while the present-equivalent branch has a lower acceleration scale and a larger curvature radius.

6 Time Branch

Table 2 shows the smooth time branch associated with the midpoint stress-test scale. The table is not a merger history and is not a reconstructed evolutionary path for an individual descendant. It only shows how the same present-equivalent branch maps into different cosmic epochs under the adopted expansion scaling.

Table 2: Smooth CPTG time branch implied by the midpoint compact-quiescent redshift stress test.

Redshift	Cosmic age	$E(z)$	$a_\star(t)$	$R_C(t)$
0.00	13.80 Gyr	1.000	6.20×10^{-10}	2.19×10^{28} m
1.00	5.85 Gyr	1.790	1.11×10^{-9}	1.22×10^{28} m
1.80	3.62 Gyr	2.757	1.71×10^{-9}	7.93×10^{27} m

The time branch makes the redshift interpretation explicit. The compact galaxy is observed on a stronger early-time acceleration branch. Once normalized to the present-equivalent scale, the branch falls in the same broad compact-quiescent range as other high-redshift effective-radius stress tests.

7 Dynamical-Mass Consistency Check

The source paper reports the following dynamical and stellar mass scales [1]:

$$M_{\rm dyn} = (1.7 \pm 0.5) \times 10^{11} M_\odot, \tag{25}$$

and

$$M_\star \simeq 1.5 \times 10^{11} M_\odot. \tag{26}$$

The ratio of the central values is

$$\frac{M_{\rm dyn}}{M_\star} \simeq 1.13. \tag{27}$$

The same midpoint observables also define a raw effective-radius mass scale:

$$M_{\sigma R} = \frac{\sigma_\star^2 R_e}{G} \simeq 3.30 \times 10^{10} M_\odot. \tag{28}$$

This is lower than the quoted dynamical mass because a pressure-supported compact galaxy is not converted from dispersion and effective radius by the unit coefficient alone. The published dynamical estimate corresponds to an effective structural virial coefficient

$$K = \frac{G M_{\rm dyn}}{\sigma_\star^2 R_e} = \frac{M_{\rm dyn}}{M_{\sigma R}} \simeq 5.16. \tag{29}$$

This coefficient is not a new CPTG parameter. It represents the ordinary geometric, projection, aperture, anisotropy, and stellar-profile factors that enter a dynamical mass estimate for a pressure-supported system.

This agreement is important for the stress test. It means the compact stellar support scale is not obviously incompatible with the published mass scale. The object is therefore a cleaner compact-quiescent redshift stress target than systems where the stellar mass, dynamical mass, and environment point to very different interpretations.

This consistency check should not be read as a CPTG mass validation. It is a sanity check on the input system. The CPTG calculation performed here tests the redshift-normalized effective-radius acceleration branch, not a full mass decomposition.

8 Uncertainty Bracket

The dominant observational uncertainty comes from the stellar velocity dispersion. The published value is [1]:

$$\sigma_\star = 294 \pm 51\,\mathrm{km\,s^{-1}}. \tag{30}$$

The effective-radius uncertainty is [1]:

$$R_e = 1.64 \pm 0.15\,\mathrm{kpc}. \tag{31}$$

Using the independent extrema gives a broad redshift-normalized range:

$$3.88 \times 10^{-10} \le a_{\sigma,z} \le 9.39 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{32}$$

Table 3 shows the central value and bounding cases.

Case	$\sigma_\star$	R_e	$a_{\sigma,z}$
Low bound	$243\,\mathrm{km\,s^{-1}}$	1.79 kpc	3.88×10^{-10}
Midpoint	$294\,\mathrm{km\,s^{-1}}$	1.64 kpc	6.20×10^{-10}
High bound	$345\,\mathrm{km\,s^{-1}}$	1.49 kpc	9.39×10^{-10}

Table 3: Effective-radius redshift stress scale for the central value and uncertainty extrema.

The midpoint value is the reference calculation. The range should be treated as an uncertainty bracket, not as a fitted interval.

9 Interpretation

NMBS-C7447 passes the compact-quiescent CPTG redshift stress test in the limited effective-radius sense. Its observed support scale at redshift

$$z = 1.80 \tag{33}$$

is high, as expected for a compact massive quiescent galaxy. After expansion normalization, however, the scale becomes

$$a_{\sigma,z} = 6.1958 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{34}$$

The key point is not that the local acceleration is small. It is not. The key point is that the high local acceleration at the emission epoch reduces to a plausible present-equivalent compact-galaxy branch when divided by the expansion factor.

The source paper argues that compact massive quiescent galaxies at high redshift were smaller, denser, and higher-dispersion than comparable present-day systems [1]. The CPTG stress-test result is consistent with that picture. The object sits on a stronger early-time acceleration branch, while its present-equivalent value remains finite and comparable to the compact high-redshift stress-test scale.

10 Data Needed for a Stronger Test

The present calculation is limited by the available scalar observables reported for NMBS-C7447 [1]. A stronger CPTG test would require spatially resolved stellar kinematics. The most important missing quantities are

$$\sigma_\star(R), \tag{35}$$

and

$$V_\star(R), \tag{36}$$

across the stellar body. A full radial test would also require a resolved mass profile, a consistent aperture correction for the measured dispersion, and a clear separation between pressure support and any ordered rotation.

With those data, the object could be tested using a proper radial Jeans reduction or a full CPTG radial field reconstruction. Without them, the appropriate classification is a compact-quiescent redshift stress test, not a full dynamical validation.

11 Technical Summary

The observed midpoint quantities are taken from the published NMBS-C7447 measurement [1]:

$$\sigma_\star = 294\,\mathrm{km\,s^{-1}}, \tag{37}$$

and

$$R_e = 1.64\,\mathrm{kpc}. \tag{38}$$

They give

$$a_\sigma = 1.7080 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{39}$$

At the observed redshift,

$$E(1.80) = 2.7568. \tag{40}$$

Therefore the redshift-normalized stress scale is

$$a_{\sigma,z} = 6.1958 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{41}$$

The corresponding curvature radii are

$$R_C(t_{\mathrm{em}}) = 7.9347 \times 10^{27}\,\mathrm{m}, \tag{42}$$

and

$$R_{C,0} = 2.1874 \times 10^{28}\,\mathrm{m}. \tag{43}$$

The stress-test outcome is positive in the limited effective-radius sense. It supports the CPTG time-scaled compact-quiescent acceleration branch, while leaving full radial validation for future resolved data.

12 Conclusion

NMBS-C7447 provides a clean second target for the compact high-redshift CPTG stress-test sequence [1]. Its measured dispersion, effective radius, dynamical mass, and stellar mass are mutually consistent with a compact massive quiescent system. The local effective-radius support scale is high, but after redshift normalization it becomes

$$a_{\sigma,z} = 6.1958 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{44}$$

This value supports the CPTG redshift stress-test branch for compact quiescent galaxies. The result should be stated carefully: CPTG passes this redshift stress test in the effective-radius sense, but the available data do not constitute a full radial dynamical validation.

References

[1] J. van de Sande, M. Kriek, M. Franx, P. G. van Dokkum, R. Bezanson, K. E. Whitaker, G. Brammer, I. Labbé, P. J. Groot, and L. Kaper, "The stellar velocity dispersion of a compact massive galaxy at z=1.80 using X-Shooter: confirmation of the evolution in the mass-size and mass-dispersion relations," Astrophysical Journal Letters, 736, L9, 2011, arXiv:1104.3860, DOI: 10.1088/2041-8205/736/1/L9.

[2] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," Astronomy & Astrophysics, 641, A6, 2020, DOI: 10.1051/0004-6361/201833910.

[3] Carter L Glass Jr, Curvature Polarization Transport Gravity: A Variational Framework for Galaxies and Cluster Mergers, DOI: 10.13140/RG.2.2.30385.95841/1.

[4] CPTG, Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity, companion resource for the present work, available at `https://github.com/CLG2025/CPTG`.